European IFR Flight Movement Analysis (2016–2022)

DASHBOARD & INSIGHTS

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BACKGROUND & USER

Background

This dashboard analyzes IFR (Instrument Flight Rules) flight movements across Europe.
It converts multi-year aviation data into clear, meaningful insights showing airport, country, seasonal, and weekly flight patterns.

Who is the User?

- Aviation analysts
- Airport authorities
- Air traffic management teams

Goal

To help users understand where flight traffic is concentrated, how it changes over time, and which regions or airports dominate European IFR operations.

DATASET OVERVIEW

Dataset Includes:

- Airport Name & ICAO Code
- Country
- IFR Arrivals
- IFR Departures
- Total IFR Movements
- Monthly & Yearly data
- Geographic coordinates (for mapping)

NEW FIELDS CREATED

1. Seasonal Categories
Mapped months into:

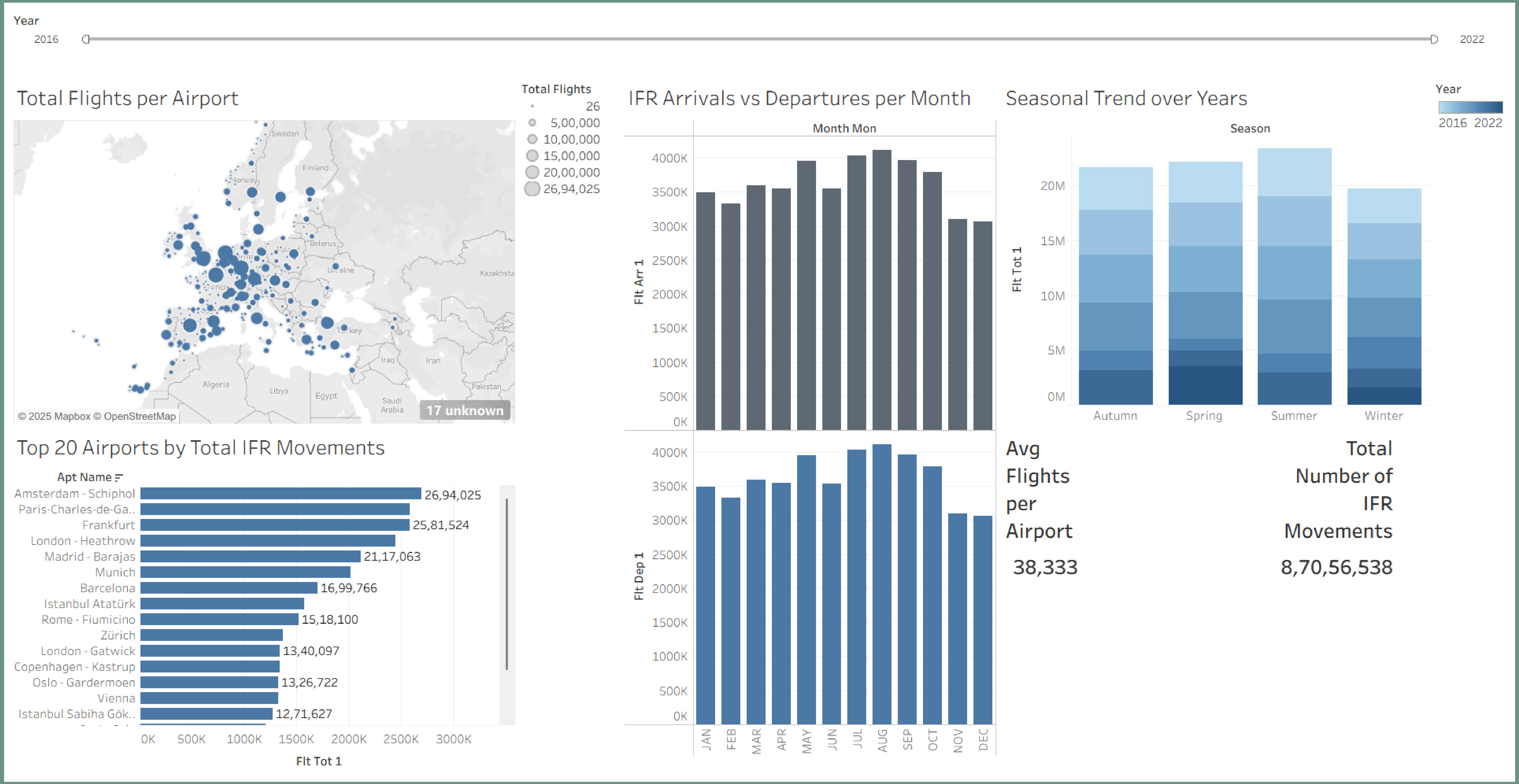
- Winter
- Spring
- Summer
- Autumn
- Used to detect seasonal travel patterns.

2. YoY % Growth
Percentage difference between each year's traffic to highlight long-term change and COVID-19 recovery trends.
3. Day-of-Week Traffic Grouping
Aggregated IFR movements for each day of the week to identify weekly peak days.

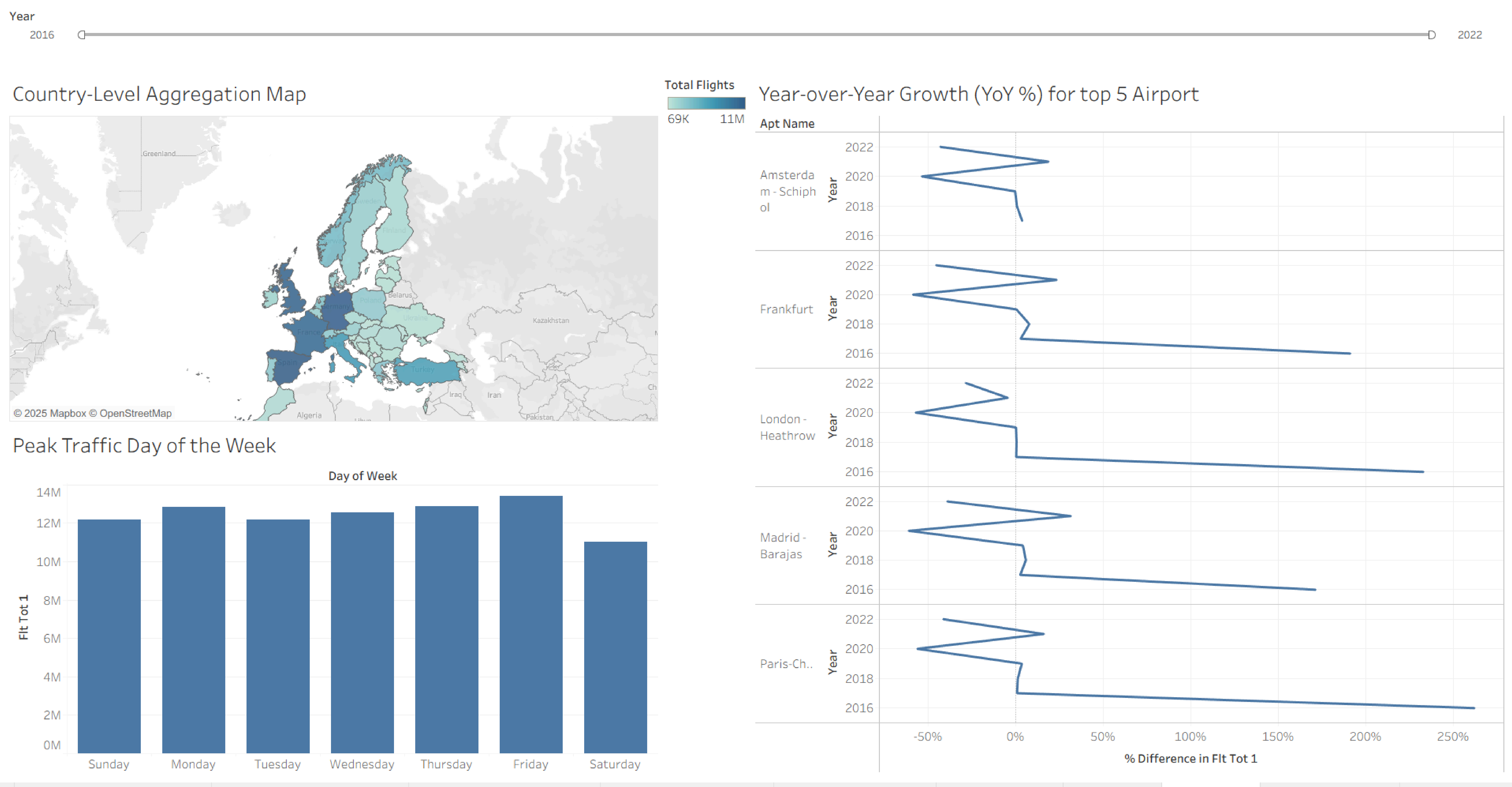
ISSUES FACED

- Large variation in airport sizes caused imbalance in visualization scales.
- Significant drop in 2020 due to COVID-19 caused extreme YoY fluctuations.
- Some airports had incomplete or inconsistent data across years.
- Mapping required geographic cleaning to ensure accurate airport placements.
- High values (millions of flights) required careful formatting for readability.

DASHBOARD


Year
2016
2022
Total Flights per Airport
Total Flights
26
5,00,000
10,00,000
15,00,000
20,00,000
26,94,025
Sweden
Finland
Norway
Belarus
Ukraine
Kazakhstan
Turkey
Iraq
Iran
Algeria
Libya
Egypt
Saudi Arabia
Pakistan
© 2025 Mapbox © OpenStreetMap
17 unknown
IFR Arrivals vs Departures per Month
Month Mon
Flt Arr 1
Flt Dep 1
4000K
3500K
3000K
2500K
2000K
1500K
1000K
500K
0K
JAN
FEB
MAR
APR
MAY
JUN
JUL
AUG
SEP
OCT
NOV
DEC
Seasonal Trend over Years
Year
2016 2022
Season
Flt Tot 1
20M
15M
10M
5M
0M
Autumn
Spring
Summer
Winter
Top 20 Airports by Total IFR Movements
Apt Name
Amsterdam - Schiphol 26,94,025
Paris-Charles-de-Ga..
Frankfurt 25,81,524
London - Heathrow
Madrid - Barajas 21,17,063
Munich
Barcelona 16,99,766
Istanbul Atatürk
Rome - Fiumicino 15,18,100
Zürich
London - Gatwick 13,40,097
Copenhagen - Kastrup
Oslo - Gardermoen 13,26,722
Vienna
Istanbul Sabiha Gök.. 12,71,627
0K
500K
1000K
1500K
2000K
2500K
3000K
Flt Tot 1
Avg Flights per Airport
38,333
Total Number of IFR Movements
8,70,56,538

DASHBOARD



Year
2016
2022
Country-Level Aggregation Map
Total Flights
69K
11M
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Greenland
Kazakhstan
Algeria
Iraq
Iran
Pakistan
Peak Traffic Day of the Week
Day of Week
14M
12M
10M
8M
6M
4M
2M
0M
Flt Tot 1
Sunday
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Year-over-Year Growth (YoY %) for top 5 Airport
Apt Name
Amsterdam - Schiphol
Frankfurt
London - Heathrow
Madrid - Barajas
Paris-Ch..
Year
2022
2020
2018
2016
-50%
0%
50%
100%
150%
200%
250%
% Difference in Flt Tot 1

LEARNINGS

- Proper data grouping (seasons, years) greatly improves clarity.
- Choosing the right chart type (bars, maps, lines) strengthens storytelling.
- Multi-view dashboards reveal patterns that single charts cannot.
- Data normalization is important when comparing airports of very different sizes.
- Mapping visuals are highly effective for geographic insights.

WHY TABLEAU?

I chose Tableau because:

- Drag-and-drop makes large aviation data easy to explore.
- Strong mapping capabilities for airport and country-level views.
- Excellent for multi-year trend analysis and seasonal comparisons.
- Smooth interactivity with filters, highlights, and tooltips.
- Clean, professional visuals ideal for analytical storytelling.

Compared to Excel or Power BI, Tableau allowed me to:

- Build advanced maps quickly
- Combine multiple time-series views
- Handle large datasets efficiently
- Create a polished and intuitive dashboard layout

WHAT I WOULD DO DIFFERENTLY

- Add more filters (country, airport size, region).
- Include forecasting for future IFR traffic trends.
- Add congestion/traffic-density heatmaps.
- Standardize airport names and codes more thoroughly.
- Improve design with color-coded themes for seasons and countries.

CONCLUSION

This project transformed raw IFR movement data into an insightful, interactive dashboard. It highlighted major aviation hubs, seasonal trends, weekly peaks, and pandemic-driven fluctuations. Through this process, I improved my skills in data cleaning, trend analysis, mapping, and dashboard design. Overall, this was a valuable hands-on experience in aviation analytics using Tableau.

The End